



B.Sc. Information Technology, FUTO

Federal University of Technology, Owerri. School of Information and Communication Technology (SICT), Department of Information Technology.

About this outline

FUTO teaches Information Technology as a five-year degree, not the four-year version most other universities run. As a university of technology, FUTO builds industrial training into the programme, so a student passes through five levels (100 to 500) and spends real time on the job before graduating.

The department sits in the School of Information and Communication Technology, which FUTO set up in 2018 to bring its computing departments together. IT sits next to Computer Science, Cyber Security, and Software Engineering there, and it used to belong to the School of Management Technology before the move.

The course content below is the national IT curriculum set by the NUC's CCMAS, which is what every Nigerian university teaches. FUTO spreads that content across five years and adds its own electives and a few extra IFT courses to fill out the programme. So a full FUTO handbook will list some courses beyond these, but everything here is part of the core, and it is safe to build your Syllabify material around it.

Each course shows its code, title, and unit load, and further down you get a short description of what the course is meant to cover. Use that as the skeleton, then write your own notes, summaries, and past questions on top of it.

How the five years run

The rhythm is coursework in the early years, industrial training in the middle and fourth year, and the project in the final year.

- **100 level (Year 1).** Foundation courses in maths, physics, and computing, plus the general studies courses.
- **200 level (Year 2).** Core programming and IT courses. A three-month industrial attachment (SIWES I) follows during the long vacation.

- **300 level (Year 3).** The bulk of the IT courses. Another three-month attachment (SIWES II) follows during the long vacation, and the student submits and defends an industrial training report.
- **400 level (Year 4).** Advanced courses run in the first semester. The second semester is given over entirely to industrial training, a four-month SIWES placement with no lecture courses. This is the main industrial training block.
- **500 level (Year 5).** The final year. Specialization courses and electives, a seminar, and the final year project, which the student builds, writes up, and defends.

To graduate, a student needs to pass all the compulsory courses and clear the minimum unit requirement, and the industrial training and final project have to be passed and defended.

Course structure

100 level (Year 1)

Code	Course title	Units	Status
GST 111	Communication in English	2	Compulsory
GST 112	Nigerian Peoples and Culture	2	Compulsory
MTH 101	Elementary Mathematics I (Algebra and Trigonometry)	2	Compulsory
MTH 102	Elementary Mathematics II (Calculus)	2	Compulsory
PHY 101	General Physics I (Mechanics)	2	Compulsory
PHY 102	General Physics II (Electricity and Magnetism)	2	Compulsory
PHY 107	General Practical Physics I	1	Compulsory
PHY 108	General Practical Physics II	1	Compulsory
STA 111	Descriptive Statistics	3	Compulsory
COS 101	Introduction to Computing Sciences	3	Compulsory
COS 102	Problem Solving	2	Compulsory

200 level (Year 2)

Code	Course title	Units	Status
GST 212	Philosophy, Logic and Human Existence	2	Compulsory
ENT 211	Entrepreneurship and Innovation	2	Compulsory
COS 201	Computer Programming I	3	Compulsory
COS 202	Computer Programming II	3	Compulsory
IFT 203	Introduction to Web Technologies	2	Compulsory
IFT 205	Introduction to Information Technology	2	Compulsory
IFT 211	Digital Logic Design	2	Compulsory
IFT 212	Computer Architecture and Organisation	2	Compulsory
INS 202	Human-Computer Interface	2	Compulsory
IFT 299	SIWES I (long-vacation industrial training)	3	Compulsory

300 level (Year 3)

Code	Course title	Units	Status
GST 312	Peace and Conflict Resolution	2	Compulsory
ENT 312	Venture Creation	2	Compulsory
IFT 302	Web Application Development	2	Compulsory
IFT 304	Web Development using Content Management Systems	2	Compulsory
IFT 308	Ethics and Legal Issues in IT	2	Compulsory
IFT 310	Mobile Application Development	2	Compulsory
IFT 322	IT Innovation and Entrepreneurship	2	Compulsory
IFT 342	Network Servers and Infrastructures	2	Compulsory
CSC 308	Operating Systems	3	Compulsory
ICT 305	Data Communications Systems and Network	3	Compulsory
IFT 399	SIWES II (long-vacation industrial training)	3	Compulsory

400 level (Year 4)

First semester runs the advanced courses below. The second semester is the four-month industrial training placement (SIWES) on its own, with no lecture courses.

Code	Course title	Units	Status
COS 409	Research Methodology and Technical Report Writing	2	Compulsory
IFT 403	Mobile and Pervasive Computing	2	Compulsory
IFT 410	System Integration and Architecture	2	Compulsory
IFT 442	Wireless Communications and Networking	2	Compulsory
INS 401	Project Management	2	Compulsory
SIW 400	Industrial Training (second semester, four months)	6	Compulsory

500 level (Year 5)

The final year carries the project across both semesters, alongside a seminar and specialization electives set by the department.

Code	Course title	Units	Status
IFT 501	Seminar	1	Compulsory
IFT 597	Final Year Project I	3	Compulsory
IFT 598	Final Year Project II	3	Compulsory
Electives	Specialization electives (chosen from the departmental list)	varies	Elective

What each course covers

100 level

GST 111, Communication in English. How English sounds and how it is built, from word classes up to full sentences. Grammar and everyday usage. Logic and clear reasoning for presentations. The writing process from first draft to proofreading, covering essays, letters, a CV, note-taking, and report writing. Reading and comprehension skills, and public speaking.

GST 112, Nigerian Peoples and Culture. Nigerian history, culture, and art up to 1800, including the Yoruba, Hausa, Igbo, and minority groups. Colonial rule and the 1914 amalgamation. How Nigeria formed as one political unit, the nationalist struggle, and the challenges of nation-building. Indigenous trade and apprenticeship, social justice, citizenship, and the values and vices that shape national life.

MTH 101, Elementary Mathematics I. Set theory, subsets, unions,

intersections, and Venn diagrams. Real numbers and their types, mathematical induction, sequences and series. Quadratic equations and the binomial theorem. Complex numbers and the Argand diagram. Trigonometric functions and identities.

MTH 102, Elementary Mathematics II. Functions of a real variable, graphs, limits, and continuity. The derivative as a rate of change and the main techniques of differentiation. Curve sketching. Integration as the reverse of differentiation, methods of integration, and using definite integrals to find areas and volumes.

PHY 101, General Physics I. Units and dimensions, vectors and scalars. Motion described through displacement, velocity, and acceleration. Newton's laws and how to apply them. Work, energy, and momentum, and the conservation laws behind them. Rotational motion, moments of inertia, and gravitation, including Kepler's laws and satellite orbits.

PHY 102, General Physics II. Electric charge, Coulomb's law, electric fields and potential, and Gauss's law. Capacitance and DC circuits with Ohm's law. Magnetic fields, the Biot-Savart and Ampere laws, and magnetic dipoles. Electromagnetic induction, Faraday and Lenz's laws, transformers, and an introduction to Maxwell's equations and AC circuits.

PHY 107, General Practical Physics I. A lab course built around careful measurement. Students take readings, tabulate data, spot common experimental errors, plot and read graphs, and draw conclusions from the numbers. The experiments back up the theory in PHY 101 and PHY 102.

PHY 108, General Practical Physics II. A continuation of PHY 107 in the second semester, with the same focus on measurement, error handling, and graphs, plus writing and presenting a proper practical report.

STA 111, Descriptive Statistics. Types and sources of data and how to collect it. Presenting data in tables, charts, and graphs. Measures of location, spread, and partition. Skewness and kurtosis. Rates, ratios, and index numbers.

COS 101, Introduction to Computing Sciences. A tour of computing: a short history, the parts of a computer, input and output devices, and the split between hardware, software, and the people who use them. How information is processed, what the internet is and does, and the different branches and jobs in computing. The lab work puts students on real machines, operating systems, and common applications.

COS 102, Problem Solving. How to break a problem down and solve it. Types of problems and the methods used on them, from algorithms and heuristics to techniques like abstraction, trial and error, and divide and conquer. Designing a solution with flowcharts, pseudocode, and decision tables, then turning it into a working program in a language such as Python or C.

200 level

GST 212, Philosophy, Logic and Human Existence. What philosophy is and its main branches, with logic as its core tool. The rules of reasoning, valid versus invalid arguments, and common fallacies. Deduction and induction. Thinking critically about texts and everyday claims, and how philosophy touches politics, religion, values, and character.

ENT 211, Entrepreneurship and Innovation. What an entrepreneur is and the theory behind risk-taking and opportunity. The traits of entrepreneurs and the place of small businesses in creating jobs and wealth. Innovation, forming and registering a business, partnerships, and joint ventures. Nigerian entrepreneurs and the support networks around them, plus the basics of e-commerce.

COS 201, Computer Programming I. The essentials of programming and the main programming styles. Data types, variables, expressions, and operators. Object-oriented ideas: classes, objects, methods, encapsulation, and inheritance. Control structures, arrays, strings, simple searching and sorting, recursion, and event handling. The lab work covers writing and tracing small programs.

COS 202, Computer Programming II. A second pass over object-oriented programming, going deeper into polymorphism, abstract classes, and interfaces. Organising code with packages, using library collections like lists, stacks, and queues, and handling events and exceptions. Building graphical user interfaces, with lab work heavy on problem-solving and program design.

IFT 203, Introduction to Web Technologies. Where the internet and the web came from, and how browsers and servers talk over HTTP and HTTPS. Building simple pages with HTML, CSS, and JavaScript, and keeping structure separate from styling. Web development tools and frameworks, valid markup, W3C standards, and adding interactive graphics and multimedia.

IFT 205, Introduction to Information Technology. What IT is as a field and where it sits next to related disciplines. Software, hardware, and networking basics, and the IT systems model. The difference between data and information, a short history of IT and the internet, and where IT gets applied. Security, privacy, and policy issues, plus a look ahead at data mining, natural language processing, and blockchain.

IFT 211, Digital Logic Design. How everything inside a computer, including instructions, is really data. Number systems and how negative and non-numeric values are represented. Boolean algebra and simplifying logic functions. Combinational circuits built from multiplexers, decoders, and adders, and sequential circuits built from flip-flops, registers, and counters. The lab work builds these circuits for real.

IFT 212, Computer Architecture and Organisation. The classic von Neumann machine and its main parts. Instruction sets and formats, memory and I/O instructions, addressing modes, the stack, and interrupts. Data path and control unit design, microprogramming, and the memory hierarchy with cache and virtual memory. The lab work includes some assembly language programming.

INS 202, Human-Computer Interface. The ideas behind designing interfaces people can actually use. Principles of graphical interfaces and the toolkits that build them. System design methods, user mental models, and human ergonomics. Evaluating and improving a design around real users, with lab work on GUI design and testing.

IFT 299, SIWES I. The first industrial attachment. Students spend about three months during the long vacation attached to a real organisation, public or private, to get hands-on experience across areas of computing. A supervisor monitors the placement, and the student keeps records, submits a report, and defends it.

300 level

GST 312, Peace and Conflict Resolution. Peace, conflict, and security in a multi-ethnic country. Types and theories of conflict, and the roots of violence in Africa, from border and political disputes to economic inequality, studied through real Nigerian cases. Peacebuilding, mediation, and the legal system, plus the roles of the UN, African Union, ECOWAS, the media, and

traditional institutions.

ENT 312, Venture Creation. The practical steps of starting a business.

Spotting opportunities and unmet needs, developing a business idea, and planning the venture. Sources of funding, from savings to venture capital.

Marketing and e-commerce, running a small or family business, negotiation, and applying newer technologies like AI, IoT, blockchain, and cloud computing to a startup.

IFT 302, Web Application Development. Building dynamic sites with a modern framework and language such as PHP or ASP.NET. The split between client-side and server-side code, and connecting a site to a database. Designing, building, testing, and publishing web applications, with hands-on work using open-source tools like Apache, PHP, Python, JavaScript, and MySQL.

IFT 304, Web Development using Content Management Systems. Building and running websites with a CMS such as Joomla or SharePoint. Setting up, deploying, and maintaining sites, and weighing CMS tools against what a client needs. Front-end and back-end work, interface and accessibility concerns, writing plugins and extensions, and turning a static design into a live CMS-driven site.

IFT 308, Ethics and Legal Issues in IT. The social, ethical, and legal side of using technology in government, business, and industry. Intellectual property, plagiarism, software piracy, copyright, patents, and trademarks. Professional codes of conduct, workplace issues like harassment and discrimination, identity theft, ethical hacking, and privacy.

IFT 310, Mobile Application Development. How mobile apps are built and the platforms they run on. Mobile operating systems, app architecture, and the main components. Building interactive apps with widgets and web services, 2D graphics, local databases, and networking. Android UI and data storage, advanced features like sensors, camera, and GPS, and measuring how an app performs.

IFT 322, IT Innovation and Entrepreneurship. Turning IT ideas into businesses. Innovation and product development, digital marketing, and the entrepreneurial openings inside IT. The startup process, feasibility and market research, business models and plans, legal and ethical issues, and a study of a real IT venture.

IFT 342, Network Servers and Infrastructures. Running network services and

servers. IPv6 addressing, DHCP and DNS, and secure links over VPNs. Virtual computing, cloud computing, and VoIP. Traffic monitoring, MPLS, and network neutrality. The lab work covers configuring IPv6 networks, working with virtual servers, and balancing and securing traffic.

CSC 308, Operating Systems. How an operating system is designed and built. Types of operating systems and their structure. Processes and threads, CPU scheduling, and synchronisation. Memory and virtual memory, file systems, and I/O. Security and protection, and a look at distributed systems. The lab work is hands-on with Linux and Unix commands.

ICT 305, Data Communications Systems and Network. How data moves between machines. Transmission types, modes, and error detection. Network protocols and the seven-layer OSI model, plus the IEEE 802 standards. Local area networks and their access methods, client-server setups, and network operating systems. TCP/IP, internet addressing, IPv4 and IPv6, and basic system administration and security.

IFT 399, SIWES II. The second industrial attachment, again about three months during the long vacation. Students deepen the practical skills they started building in SIWES I, keep monitored records, and submit and defend an industrial training report.

400 level

COS 409, Research Methodology and Technical Report Writing. How to run and write up research. Types and approaches to research and the research process. Formulating a problem, writing a proposal, and reviewing the literature. Collecting data through questionnaires, interviews, and ethnography, and building good data instruments. Writing a technical report, citing sources, and preparing references. This course sets a student up for the final year project.

IFT 403, Mobile and Pervasive Computing. Programming for mobile and everywhere-present devices. The theory behind pervasive computing, context-aware interaction, and working within device limits. Building systems from sensors, actuators, and embedded parts. Capturing requirements for pervasive systems, sensor networks and their security, and an introduction to cloud services.

IFT 410, System Integration and Architecture. Bringing separate systems

together as part of the development lifecycle. Integration architecture, methods, and technologies. System testing, evaluation, and benchmarking, contracts and RFPs, deployment and release, and acceptance testing. DevOps principles, enterprise architecture, and how rolling out IT interacts with an organisation's own processes.

IFT 442, Wireless Communications and Networking. The principles behind wireless data. Radio propagation, antennas, digital modulation, and spread spectrum, with standards like Wi-Fi, WiMAX, and Bluetooth. Designing, testing, and deploying wireless networks. Cellular concepts, frequency reuse, GSM, UMTS, and LTE, mobility management, Mobile IP, and personal area networks.

INS 401, Project Management. Planning and running a project from start to finish. The project management lifecycle and its processes. Managing teams, communication, scope, scheduling, resources, quality, and risk. Handling procurement, whether in-house, outsourced, or offshore. Executing, controlling, and closing a project, including cost and change control and project auditing.

SIW 400, Industrial Training. The main industrial training block, a four-month placement that fills the whole second semester. The student works inside a real organisation, applies what the first four years taught, keeps a logbook, and submits and defends an industrial training report at the end.

500 level

IFT 501, Seminar. A supervised seminar. The student researches a current IT topic, writes it up, and presents it to staff and classmates. This builds the research and presentation skills the final project needs.

IFT 597, Final Year Project I. The first half of the final-year project. Working under a supervisor, the student picks an IT problem, submits a proposal, reviews the literature, and settles on a method. The introduction, literature review, and methodology are submitted for grading at the end of the first semester.

IFT 598, Final Year Project II. The second half, carrying the project through to the end. This is where the system is built and evaluated. The student finishes the remaining chapters, pulls the full report together, and defends it in an oral presentation before the department.

Specialization electives. In the final year the department offers elective

courses that let a student go deeper into an area such as data science, networking and security, or software systems. The exact list is set by the department each session.

Facilities the programme is meant to have

The NUC expects an IT department to run three kinds of laboratory, and FUTO's IT department moved into a new, equipped building in 2023. The three labs are:

- A software lab with enough machines for every student in a lab-based class, loaded with compilers, debuggers, operating systems, and design tools such as UML. The recommended ratio is about one staff member to every 25 students in practical sessions.
- A digital logic lab stocked with logic gates, multiplexers, flip-flops, logic boards, an oscilloscope, and memory chips for the hardware courses.
- A network lab with routers, switches, cabling and crimping tools, and network software like Cisco Packet Tracer, GNS3, Wireshark, and PRTG, so students can practise the net-centric courses.

On staffing, the guideline is one academic staff member for every 20 students, with at least six academic staff needed to run the programme and most of them holding a PhD. The department also needs its own library access, lecture rooms, a conference room, a seminar room, and offices sized to NUC space standards.

A note on accuracy

The five-year shape of the programme, the industrial training pattern, and the final-year project are how FUTO actually runs the degree, confirmed against the university's own departmental pages. The course descriptions come straight from the national CCMAS curriculum. The one place to double-check against a current departmental handbook is the exact level and semester each course sits in, and the codes and unit loads for the 400-level industrial training, the 500-level seminar, the project, and the specialization electives, since FUTO sets those itself and can revise them from session to session.

Sources

- FUTO, School of Information and Communication Technology and Department of

Information Technology: <https://futo.edu.ng/school-of-information-and-communication-technology-sict/departments-of-information-technology/>

- FUTO five-year structure and SIWES pattern, from the university's departmental pages at futo.edu.ng.
- NUC CCMAS, Computing discipline (B.Sc. Information Technology), 2023. Local copy in `docs/ccmas/disciplines/Computing-CCMAS.pdf` . The full, unedited course contents are in `docs/course-outlines/information-technology.md` .